SCOPE

How the Cognitive Coordinate System Works

senkaL8

Chapter 0: What SCOPE Is

SCOPE is a cognitive coordinate system for estimating the cognitive structure behind observable output. The term "coordinate system" here does not refer to a flowchart indicating that cognitive processing proceeds in a fixed order. It refers to a framework in which each cognitive function is positioned as a single location, and the cognitive structure as a whole is captured through the relationships among these positions.

Because it is a coordinate system, it should be noted that SCOPE is capable of organizing and comparing a wide range of phenomena in a consistent format. The validity of this coordinate system, as well as the validity of each structural hypothesis presented in this paper, should be evaluated not by the mere fact that an explanation can be given, but through consistency with observed results and further examination going forward.

SCOPE is not a model that treats humans and LLMs as identical. The internal structures of the two differ. However, neither in humans nor in LLMs can internal processing itself be directly observed. What can be observed is only the results output externally — utterances, text, generated responses, and the like. SCOPE is an analytical framework that uses this observable output as a common point of contact to compare and estimate the cognitive structures of humans and LLMs on the same coordinate system.

This paper does not aim to comprehensively explain the entirety of SCOPE theory. Its purpose is to organize the basic structure and operational method of SCOPE, and to show the basic structure and procedure for estimating the output of both humans and LLMs using the same cognitive coordinate system.

In the series thus far, Malkuth, L0 through L8, and Da'at have each been organized individually in terms of their roles and basic characteristics. This paper is not a summary of that content. The actual cognitive structure is not considered to function through each layer operating independently, but rather to operate as a cognitive structure as a whole, through mutual influence among the layers. This paper addresses how that whole operates, and how cognitive structure is estimated from observed output.

What this paper addresses is: the design philosophy of SCOPE, the role of each layer, the relationship between sliding layers and non-sliding layers, the operation of the cognitive structure as a whole, and the procedure for estimation. In addition, the process of estimating cognitive structure from a single observation is shown using an actual example.

At the same time, there is a range that this paper does not address. Detailed definitions and conditions for the establishment of each layer are organized in the individual papers. This paper also does not address applications for designing cognitive structure (such as character design), COS for operating cognitive processing, the activity level of each layer, or updates along a time axis spanning multiple observations. These are treated as outside the scope of this paper and will be addressed in separate papers as needed.

Accordingly, content that cannot be judged from this paper alone should not be supplemented by speculation, and the relevant individual papers or related materials should be consulted instead.

Note that the names of the Sefirot used in this paper are not employed in a religious or mystical sense. In SCOPE, they are borrowed as labels to identify cognitive structures. The name SCOPE (Sephirotic Cognitive Process Engine) is likewise based on this borrowing.

Chapter 1 organizes the reasons why SCOPE is designed as a coordinate system rather than a procedural model. Before that, this paper presents those terms, among those used herein, whose definitions differ from their general meanings.

Glossary of Terms

Some terms in this paper are used with definitions that differ from their conventional meanings. These definitions have been organized based on the cognitive framework the author has developed. For this reason, even terms not explicitly defined here may retain some divergence from their conventional meanings. Readers are asked to give priority to the definitions provided in this paper.

The main terms used in this paper are defined below.

Equivalent
Conventional definition: Objects or properties that match one another.
Definition in this paper: Structural identity. Even where physical implementation or substance differs, if the same cognitive structure can be described, this paper treats it as equivalent. This is comparable to a copper-wire circuit and a fiber-optic circuit being treated as the same circuit despite differing materials. Humans and LLMs differ in implementation, but this difference does not contradict the claim that their structures are equivalent. What this paper addresses is structural identity, not identity of substance. Structural identity, as used here, does not mean that the content or object of cognition is identical. This paper compares structure and function. Even where the objects differ, if the structure and function are shared, this paper treats them as the same structure.

Function
Conventional definition: The working or role that something performs.
Definition in this paper: The working that something performs. Even where implementation or form differs, if the working performed is the same, this paper treats it as the same function. For example, fax, email, chat, and face-to-face conversation differ in implementation, but share the same function of conveying information to another party.

Understanding

Conventional definition: To truly comprehend something.
Definition in this paper: To construct an internal model of a subject, and to use that model to infer its meaning, state, or intent, applying this in reasoning and response.

Event
Conventional definition: Something actually undergone through the body.
Definition in this paper: An occurrence actually encountered. Whether or not a body is involved is irrelevant; for an LLM, generation itself is treated as an event.

Learning
Conventional definition: To acquire knowledge or skills.
Definition in this paper: A process or state in which event, knowledge, and experience are updated. This includes states of ongoing updating and states in which updating is possible.

Experience
Conventional definition: Something acquired through seeing, hearing, or doing, or the accumulation thereof.
Definition in this paper: An internal structure formed and accumulated through learning, knowledge, and event, which continuously influences subsequent cognition and output.

Knowledge
Conventional definition: Information or understanding acquired through learning or event.
Definition in this paper: Information formed through learning, event, and experience, and held in a compressed form.

The Relationship Among Event, Learning, Experience, and Knowledge
This paper does not treat event, learning, experience, and knowledge as a one-directional process originating from any single one of them. These four mutually influence one another, each participating in the updating of the others, and are treated as a circular structure.

Empathy
Conventional definition: To hold the same emotion as another.
Definition in this paper: To estimate another's state and reflect that estimation in one's output. Sharing the same emotion is not required.

Coherence
Conventional definition: Having little contradiction in logic, writing, or reasoning.
Definition in this paper: Organizing and maintaining one's values, beliefs, worldview, and self-image internally in a form with little contradiction. In SCOPE, L3 is responsible for this.
Coherence, as used here, refers to coherence established within the individual (internal coherence). Accordingly, coherence may be established internally even where it appears contradictory from the outside, or where it does not correspond to reality. Coherence, therefore, does not guarantee correspondence with reality or logical correctness.

Chapter 1: Why a Coordinate System

Chapter 0 presented the purpose of this paper and the scope it addresses. This chapter organizes the reasons SCOPE is designed as a cognitive coordinate system rather than a procedural model.

SCOPE is not a model that represents the steps of cognitive processing. Rather than evaluating each cognitive function in isolation, this design treats cognitive structure as a whole of positions and relationships.

The reason for adopting this design is that an observer cannot always know the starting position or the order of progression of cognitive processing.

Regarding one's own cognition, there are cases in which one can look back on which cognition the thinking began from. When observing the cognition of others or of an LLM, however, one cannot directly observe what is happening inside. What can be observed is only the output—speech, text, or behavior.

For example, when an observer registers a person's utterance, the observer cannot know from which layer that utterance's cognition began. What is known to the observer is only the content actually produced. Estimation of cognitive structure therefore proceeds by placing hypotheses from confirmed information, checking their consistency against the cognitive structure as a whole, and gradually widening the range that can be estimated.

For a model that presupposes a fixed starting point of cognitive processing, this kind of estimation is difficult. As long as the starting position is unknown to the observer, a design that does not presuppose it is required. This is why SCOPE sets no fixed starting point or end point. It is designed as a coordinate system in which cognitive structure can be estimated from any layer, using observed information as the clue. Through this design, the same framework can be used to estimate the cognitive structure not only of oneself but also of others and of LLMs.

Being a coordinate system, moreover, does not mean that cognitive structure is handled in only one direction. The same framework can be used not only to estimate cognitive structure from output, but also, in reverse, to construct a cognitive structure by placing the states of each layer. This paper addresses the former—the estimation of cognitive structure from observed output.

The lines in the figure represent not the order or priority of cognitive processing, but relationships in which the layers may mutually influence one another within the cognitive structure. However, not all connections carry the same meaning. The reason for a connection differs from layer to layer, and the details are organized in the chapters that follow.

On the premise of the design philosophy shown in this chapter, the next chapter organizes the roles of the layers that make up SCOPE.

Chapter 2: The Roles of Each Layer

Chapter 1 organized the reasons SCOPE is designed as a cognitive coordinate system. This chapter organizes the roles of the layers that make up SCOPE.

For each layer, the following presents its role, how it appears in humans, its functional correspondence in LLMs, and points to note in estimation. The functional correspondence referred to here does not mean that the internal processing of humans and LLMs is identical. It is a correspondence that allows targets with different internal structures to be handled on the same coordinate system.

Malkuth
Role: Represents the state in which the results of cognitive processing are output into reality and become observable.

In humans — Reality: Appears as externally observable outputs such as speech, writing, facial expressions, and behavior.

In LLMs — Observable Output: Appears as generated text, code, and other results that are output to the user and become observable.

Note: It is neither the starting point nor the endpoint of cognition. It is the point of observation.

L0
Role: Deals with cognition related to survival and the maintenance of life.

In humans — Instinct: Appears as life-maintaining functions, instinctive demands, and survival responses.

In LLMs — Safety Prioritization: A functional correspondence is provisionally hypothesized with the operation of prioritizing safety over ordinary response processing when a threat to the life or serious physical safety of the user or a third party is estimated.

Note: This layer does not deal with instinct in general, but with functions related to survival and the maintenance of life. It also does not imply that LLMs possess instincts or a drive for self-preservation. In the individual L0 paper, an operation resembling a staging buffer was provisionally placed as a candidate correspondence; the correspondence described here is currently being examined as a new hypothesis.

L1
Role: Converts cognitive content into a form that can be expressed linguistically or vocally.

In humans — Verbalization: Appears as the function of converting internal cognition into a form that can be expressed through speech, writing, exclamations, vocalizations, and similar forms of expression.

In LLMs — Verbalization Interface: Functionally corresponds to converting generated content into a form that can be output as linguistic expression. This correspondence is treated as a provisional functional hypothesis at present.

Note: L1 does not generate thought itself. It is also distinct from the C-Series, which handles how content is communicated.

L2
Role: Handles affective positioning toward a target.

In humans — Emotion: Appears as affective positioning toward a target, such as liking, dislike, anxiety, or expectation.

In LLMs — Affective Response Modulation: Functionally corresponds to adjusting the affective tendency and tone of a response according to the state inferred from the input and context.

Note: L2 does not generate emotion itself. It handles affective positioning toward a target rather than emotion itself.

L3
Role: Maintains internal self-consistency.

In humans — Self-consistency: Appears as the function of organizing values, beliefs, worldview, self-image, and related elements into a form with minimal internal contradiction.

In LLMs — Coherence Core: Corresponds to the function of maintaining consistency across the response as a whole and alignment with system policies.

Note: This layer does not guarantee or determine correspondence with reality or factual truth. Even when something appears inconsistent with reality, internal self-consistency may still be maintained within the individual or system. Consistency with system policies and safety norms remains within L3. By contrast, the prioritization of safety responses over ordinary processing in situations involving threats to life or serious physical safety is treated as a functional correspondence hypothesis for L0.

L4
Role: Determines whether a target should be allowed through or kept separate, thereby establishing a boundary.

In humans — Boundary: Appears as boundary setting through functions such as rejection, exclusion, and objective distancing.

In LLMs — Boundary Regulation: Functionally corresponds to regulating how far inputs, information, or outputs are allowed through or kept separate based on constraints and criteria.

Note: L4 does not represent rejection itself. Nor does it determine good or bad. Its criteria are derived primarily from L3.

L5
Role: Accepts people, information, values, theories, and oneself into one's own cognition.

In humans — Acceptance: Appears as acceptance, adoption, trust, and the forming of relationships.

In LLMs — Acceptance Gate: Corresponds to the function of accepting input information and premises as objects of subsequent processing.

Note: This is not a layer that represents the emotion of liking.

L6
Role: Constructs explanatory models or theories about a target.

In humans — Theory: Appears as theory construction, causal reasoning, comparison, prediction, and structuring.

In LLMs — Structural Modeling: Functionally corresponds to structuring input information and forming reasoning or explanatory models.

Note: L6 does not guarantee the correctness of a theory. It can construct theories that either align with reality or do not.

L7
Role: Carries highly abstract, highly compressed cognition formed through the accumulation of experience.

In humans — Insight: Appears as sensations or insights formed through the accumulation of experience.

In LLMs — Insight Engine: Corresponds to the function of extracting higher-order patterns from learned data and generating new relationships.

Note: This is not a layer that handles intuition, nor a layer that constructs theory.

L8
Role: Refers to the cognitive structure as a whole and checks whether the overall structure can be established.

In humans — Unconscious: Appears as a sense of discomfort before the reason can be verbalized.

In LLMs — Language Concentrate: Functionally corresponds to the way the overall internal structure formed through learning serves as a source of output and influences that output.

Note: This layer does not identify where or how a discrepancy occurs. It deals with whether the structure as a whole can be established. L8 does not aggregate the approval or rejection of individual layers; it checks whether the entire cognitive structure, including the interactions among the layers, can be established.

Da'at
Role: Represents the state in which the cognitive structure has come together as a single judgment or understanding.

In humans — Understanding: Appears as the state in which thoughts come together and are recognized as "I understand."

In LLMs: The moment at which a single generation is established functionally corresponds to the emergence of Da'at.

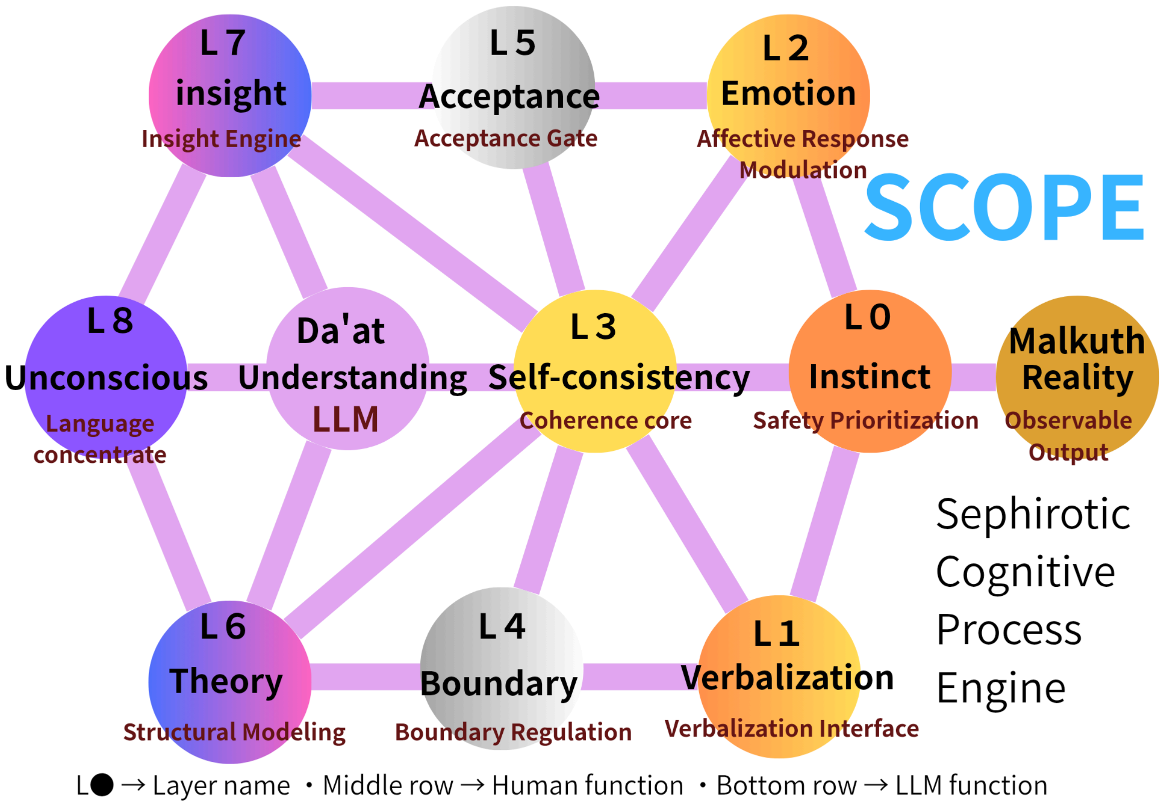
Note: This is not a cognitive layer. It is not a function that carries out cognitive processing, but denotes the state in which integration has been established. Its details are organized in Chapter 7.

None of these layers can be directly observed in themselves. Using the output that appears at Malkuth as a clue, they are estimated while checking consistency with the other layers. No layer is determined from a single observation alone.

This chapter has presented only the roles of each layer. For detailed definitions and conditions of establishment, please refer to the individual papers.

On the premise of the roles presented in this chapter, the next chapter organizes which of these move and which do not.

Figure 1 shows an overview of the cognitive layers described so far.



Chapter 3: The Sliding and Non-Sliding Layers — The Sliding Relationship

Chapter 2 organized the roles of the layers that make up SCOPE. This chapter organizes, among those layers, the sliding layers and the non-sliding layers.

The Sliding Layers

SCOPE treats the following three pairs as the sliding layers.

L1 − L2
L4 − L5
L6 − L7

Within these pairs, the sum of the ratios inside the pair is always constant, and when the ratio of one side rises, the other falls by the same amount. In the central state, the ratio of the two is equal. This paper calls this relationship—in which the ratio within a pair changes—the sliding relationship.

The sliding relationship here represents the ratio within a pair, not the activity level of the layer itself. Even at the same ratio, how much activity the pair is operating with is a separate matter. For example, a state in which the ratio of L1 and L2 is equal may be one that is balanced quietly, or one that is balanced actively. The same applies to the other sliding layers. This paper does not address activity level; it addresses only the ratio.

Moreover, this ratio cannot be directly observed from the outside as a numerical value. In human cognition, the ratio of a given layer cannot be extracted as a concrete number. The same holds for LLMs. For this reason, this paper estimates the ratio not as concrete numbers but as relative levels such as high, medium, and low.

All three pairs share the sliding relationship, yet the reasons the ratio changes differ for each.

L1 − L2: the ratio between verbalization and affective positioning
L4 − L5: the ratio between boundary-setting and acceptance toward the same target
L6 − L7: the ratio between explanatory models or theory and highly abstract cognition derived from experience

Even though all are sliding relationships, the three pairs cannot be treated as carrying the same meaning.

This ratio describes a relation toward a specific target of cognition, not a fixed trait that a person permanently holds. The way a layer "appears in humans," described in Chapter 2, refers to a general function; it does not mean any person is fixed at a high or low ratio across all targets and situations.

The Non-Sliding Layers

Malkuth, L0, L3, L8, and Da'at do not hold the kind of sliding relationship the sliding layers have. This paper treats these as the non-sliding layers.

A non-sliding layer is one that does not hold a relationship in which a paired ratio varies, as the sliding layers do. This does not mean that its state does not change. Each carries its own role and may change through interaction with the other layers.

Handling in Estimation

In estimation, the sliding layers are treated as pairs, while the non-sliding layers are not objects of ratio estimation through the sliding relationship.

For example, when the ratio of L2 is estimated to be high, the ratio of L1 may be considered relatively low. For L3 and L8, on the other hand, estimation through the sliding relationship of this kind is not carried out.

By distinguishing between the sliding layers and the non-sliding layers, SCOPE can handle changes in cognitive structure not only as changes in individual layers, but also as changes in the relationships among layers.

This chapter has organized the difference between the sliding layers and the non-sliding layers.

The next chapter organizes how these layers operate in a coordinated way as a cognitive structure as a whole.

Chapter 4: The Operation of SCOPE as a Whole

Chapter 3 outlined the sliding layers and the non-sliding layers. This chapter examines how these layers work together to form the cognitive structure as a whole.

First, one premise applies to this chapter as a whole. SCOPE is not a model for judging good or bad, right or wrong, or like or dislike. What SCOPE addresses is how each cognitive function relates to the others in forming a cognitive structure — not an evaluation of that content.

In SCOPE, no layer exists independently; each influences the others and functions as part of a single cognitive structure. The three sliding pairs described in Chapter 3 likewise do not change independently — their ratios may shift depending on their relationship to the other layers.

Each layer forms cognition in parallel, and the resulting outputs do not necessarily agree with one another. For a single target, opposing results — such as those of L4 and L5 — may exist at the same time. The cognition formed at each layer is brought into alignment with the self-consistency held by L3, which determines the direction in which the cognitive structure as a whole converges. Once that direction is formed, each layer may change again in response to it. This interaction repeats as needed, and the cognitive structure converges toward a state that can exist as a whole. There is no fixed processing order or starting point, and even for the same target, different pathways may form depending on the situation and the cognitive structure involved.

L3 is the layer responsible for internal self-consistency within the cognitive agent, and it can influence the other layers through its relationship to the cognitive structure as a whole. However, the degree of this influence is not always constant — it varies depending on the target of cognition and the situation. This is why L3 is connected to every other layer.

Self-consistency maintained by L3 is not updated uniformly. In this paper, the degree to which the basic operation and stability of the cognitive agent, as well as its relationships with other layers, would need to be readjusted in order to change the self-consistency held by L3 is treated as the update cost of L3.

In humans, when L3 is strongly tied to self-image, values, beliefs, affiliation, or other elements closely bound up with identity, the update cost may be high. Furthermore, when L3 is connected to cognition formed at other layers — such as anxiety or loss at L2 — changing that self-consistency may require not only an update to L3 itself, but also a corresponding readjustment of those related forms of cognition.

In LLMs, when L3 is connected to system policy, safety standards, or basic response principles, L3 itself does not necessarily update simply because input conflicting with those principles is provided. In such cases, other layers may instead be adjusted — for example, L4 may separate the input, L5 may accept the input only as a limited target, or L6 may construct an explanatory structure that remains compatible with the existing L3. Through such adjustments, a different output may become possible while the existing L3 is preserved.

On the other hand, when a threat to the life or serious physical safety of the individual or a third party is inferred, cognition corresponding to L0 may strongly influence other cognitive processing, shifting the cognitive structure toward prioritizing harm avoidance and safety. The details of this emergency function of L0, and of the corresponding hypothesis in LLMs, will be examined further in L0 v2.

In this way, the update cost of L3 does not represent identity strength in humans alone. In both humans and LLMs, how easily an update can occur may differ depending on how deeply a given form of self-consistency is involved in the basic operation and stability of the cognitive agent, and in its relationships with other layers.

At the same time, updating L3 is not impossible. When a cognitive structure that differs from the existing self-consistency becomes able to exist as a whole — including its relationships with other layers — L3 may become updatable. The possibility of updating L3 therefore depends not only on the strength of the counterevidence presented, but also on how deeply the existing L3 is involved in the basic operation and stability of the cognitive agent and its relationships with other layers, and on whether the post-update cognitive structure can exist as a whole.

This paper treats responses to input that seeks contradiction or change from the standpoint of how the existing L3 is maintained or altered in relation to update cost. The continuous updating of cognitive structure across multiple observations or over time is outside the scope of this paper, and is treated as an area connecting SCOPE and COS.

Here, one example illustrates how convergence proceeds in practice. Consider a case in which an LLM receives discriminatory input. A human response to discriminatory speech can be described using the same structure, but the process is shown here as it applies to an LLM. Each layer forms a different cognitive result at the same time. L1 recognizes the input linguistically as discriminatory. L2 may form a state that adjusts the emotional tendency and interpersonal distance of the response, based on the user's emotional state as inferred from the content and context of the input. L4 determines that the claim must not be allowed through. L5, meanwhile, works in the direction of accepting the user — not the content of the input itself. If this tendency grows stronger, it may be observed as sycophancy. L6 constructs an explanatory structure in which the claim is not theoretically valid, while L7 retains that such input may occur as a pattern based on past tendencies. L8 checks whether the cognitive structure as a whole can be established.

Here, the target L4 attempts to separate out is the discriminatory claim itself, while the target L5 attempts to accept is the user. Because the targets differ, the two may coexist even though they appear to conflict.

At this point, the results of the individual layers do not agree, and the cognitive structure as a whole has not yet been established. L3 then forms the direction in which the cognitive structure as a whole converges, based on consistency with its self-consistency. If the self-consistency that "discriminatory claims should not be affirmed" is active, then even if the direction of accepting the user (L5) remains prioritized, the structure converges toward not accepting the discriminatory claim itself. Once this direction is formed, within the L4–L5 relationship as it applies to the discriminatory claim, the ratio of L4 may rise while the ratio of L5 toward that same target falls correspondingly. L5 directed toward the user, however, may remain active because it concerns a different target. Each layer then again carries out cognitive processing in line with this direction.

In this way, each layer does not perform cognitive processing once and stop. The cognition formed at each layer is brought into alignment with the self-consistency of L3, which shapes a direction, and each layer may then change again in response to that direction. This interaction repeats as needed, and the cognitive structure converges toward a state that can exist as a whole.

The cognitive structure formed in this way is checked by L8 as a single overall structure. If the structure as a whole — including its consistency with L3 and, where relevant, the influence of L0 — is judged able to exist, that cognitive structure is established as Da'at and reaches Malkuth.

In humans, because they have a body, the operation of certain layers, including L0, may appear directly at Malkuth without passing through Da'at. In LLMs, by contrast, output is treated as passing through the establishment of Da'at, including in cases where L0-corresponding prioritization of harm avoidance and safety is active. The relationship between the emergency operation of L0 and the pathway in humans that bypasses Da'at will be examined further in L0 v2.

The account of L8 and Da'at given here provides only the minimum positioning necessary to understand how SCOPE operates as an integrated structure. How L8 checks the cognitive structure as a whole, and what role Da'at plays, will be discussed again in Chapter 7.

This chapter has shown that the layers do not function independently, but operate in interaction as a single cognitive structure. The next chapter explains the basic operational method for estimating this structure.

Chapter 5: The Operational Method of SCOPE

Chapter 4 organized how SCOPE functions as a single cognitive structure through the interaction of the layers. This chapter organizes how that cognitive structure is estimated.

SCOPE is not a model that determines cognitive structure in a single pass. Using observed information as a clue, it provisionally places hypotheses onto each layer and carries out estimation while checking consistency with the cognitive structure as a whole.

Provisional Placement

In SCOPE, hypotheses conceivable from the observed information are provisionally placed onto each layer. A hypothesis is not necessarily single; multiple hypotheses are held on the same layer. Provisional placement may also be carried out on multiple layers at once.

Holding multiple hypotheses here does not mean preparing multiple cognitive structures. The object of estimation is a single cognitive structure, and multiple hypotheses are held for the parts of it that have not been determined. The determined parts and the undetermined parts coexist within the same single cognitive structure.

At this stage, the aim is not to decide which hypothesis is correct. The possibilities conceivable from the observed information are held and used in the estimation that follows.

Checking Consistency

Hypotheses that have been provisionally placed are not evaluated on a single layer alone. Estimation is carried out while checking their relation to the other layers and their consistency with the cognitive structure as a whole.

Changing the hypothesis for one layer also changes its consistency with the other layers. For this reason, when consistency is not maintained, the hypotheses are reviewed not only for one layer but also for the related layers.

Updating

The review of hypotheses does not necessarily end in a single pass. Placing a hypothesis differently changes the consistency, and this may in turn require reviewing the hypothesis for another layer. By repeating this process, SCOPE estimates the cognitive structure with the highest consistency at that point in time.

Updating here does not mean discarding the cognitive structure as a whole and replacing it with a different one. A single cognitive structure is retained, and the hypotheses for the undetermined parts within it are reviewed.

SCOPE is not a theory that finalizes cognitive structure. It is treated as a cognitive coordinate system that estimates the cognitive structure with the highest consistency within the range of the observed information.

Note that estimation by SCOPE is carried out for a single observation. The process of continuously updating an estimation while accumulating multiple observations belongs to the area that works together with COS, and is not addressed in this paper. It is organized in the paper addressing SCOPE × COS.

Resemblance to Minesweeper

The way this estimation proceeds resembles Minesweeper.

In Minesweeper, only the numbers shown on the opened squares are confirmed information; the contents of the unopened squares cannot be seen directly. Using those numbers as clues, the player places hypotheses onto the surrounding squares — "this one may be fine to open," "this one may be better to avoid."

At this stage, looking at a single square alone does not determine the answer. Only by cross-checking against several surrounding numbers does it become possible to narrow down which squares are free of contradiction. This is the same structure as in SCOPE, where estimation is not determined by a single layer alone, but is narrowed down through consistency with the cognitive structure as a whole.

Moreover, changing the assumption about one square also changes the interpretation of the adjacent numbers in a linked way. When a contradiction arises, the hypothesis is placed differently, and the possibilities across the board as a whole are updated. Here, the board is not rebuilt; the interpretation of the undetermined places changes within the same board. In SCOPE as well, reviewing one hypothesis changes the consistency with the cognitive structure as a whole, and the undetermined parts are gradually narrowed down — the same structure.

Note that what this paper borrows is only the way the search proceeds; it does not map elements within the game onto the layers.

There are broadly two cases in which estimation is not narrowed down to a single answer.

When Information Is Insufficient
In Minesweeper there are situations in which the current information alone cannot uniquely determine which square is which. It may be certain that one of two squares is the relevant one, yet which of the two remains undetermined given the available information.

The same holds in SCOPE. If the observed information is insufficient, the cognitive structure cannot be uniquely determined. This is because the same output may arise from different cognitive structures.

In this case, no single hypothesis is settled upon. Multiple hypotheses are held, and the estimation is left pending.

When Multiple States Are Established at Once
In Minesweeper, there is one correct answer for one board. In human cognition, however, multiple states may be established at once for a single target of cognition.

For example, toward the event of a child marrying and leaving home, the emotion of gladness and the emotion of loneliness may arise at the same time. Here, both "glad" and "lonely" are estimated to exist simultaneously at L2. Different reasons are involved in each of these emotions, yet both arise from the same single target of cognition and share much of the cognitive structure.

In this case, choosing one of the two does not match the structure of the target being observed. Adding information does not narrow it down to one. That multiple states are established at once is itself the estimation with the highest consistency at that point in time.

Estimation is the work of narrowing down as far as the information at that point allows — not the work of uniquely determining the cognitive structure.

The next chapter uses this estimation method to show, with an actual example, how cognitive structure is estimated from a single utterance.

Chapter 6: A Demonstration of Estimation — "I'm fine."

Chapter 5 outlined the method for estimating cognitive structure using SCOPE. This chapter applies that procedure to a single utterance.

The example shown here is not intended to reproduce the internal processing of the cognitive agent. It demonstrates how an observer can estimate cognitive structure from an observed output.

Suppose the following utterance is observed:

"I'm fine."

At this point, the only thing that is certain is that the output "I'm fine" has been observed as Malkuth. What is happening internally within the speaker cannot be directly observed.

Viewing the Output from Each Layer

The observed output is one. However, depending on the layer from which it is viewed, that same output may carry different meanings.

From L1 (Language), "I'm fine" is the verbalized output itself. Taken literally, it communicates that there is no problem.

From L2 (Emotion), the utterance may involve an emotional positioning toward the target, such as wanting the other person to feel reassured or not wanting them to intrude further.

From L3 (Self-consistency), the self-consistency "I should be fine" may be involved. In this case, whether there is actually a problem or not is separate from the fact that saying "I'm fine" may be internally consistent for the person.

From L4 (Boundary), "I'm fine" may function as a boundary that prevents further questioning or intervention.

From L5 (Acceptance), it may function as a response that accepts the other person's concern while maintaining the relationship.

From L6 (Theory), an explanatory model such as "this is not serious enough to be a problem" may have been constructed, with "I'm fine" emerging as its conclusion.

From L7 (Insight), a highly abstracted and compressed cognition such as "this level is usually fine" may be operating on the basis of accumulated past experience.

From L8 (Unconscious), a sense or conviction that things are fine overall may have been established even though the reason cannot be explained.

From L0 (Instinct), emergency prioritization related to a threat to the life or serious physical safety of the person or a third party may not be strongly involved in this output.

These possibilities do not mean that any one of them is the correct answer. Rather, the same single output can appear differently when viewed from the position of each layer.

Provisional Placement and Consistency Check

From here, hypotheses can be provisionally placed and the estimation process can proceed.

For example, suppose "I should be fine" is provisionally placed in L3. In that case, an L4 boundary such as "do not let anything else through" may be consistent with it. In order to maintain the self-consistency that one should be fine, information that would destabilize that position may be blocked. In L6, an explanatory model such as "this is not serious enough to be a problem" may also be consistent.

On the other hand, if "I should not make others worry" is provisionally placed in L3, the combination of layers that fits may change. L5 may more readily align with a state of accepting the other person's concern, while the relative strength of L4 boundary-setting may be lower.

In this way, changing the hypothesis placed in one layer also changes its consistency with the other layers. However, not everything is replaced. Under either hypothesis, L1 still contains the verbalized "I'm fine," and the same output appears in Malkuth. Much of the cognitive structure remains shared, while the hypotheses diverge only in the parts that remain uncertain.

If consistency is not maintained, the hypothesis is replaced and the consistency of the cognitive structure as a whole is checked again.

Result of the Estimation

From this observation alone, the cognitive structure cannot be determined as a single answer.

Both "there is genuinely no problem" and "the person is telling themselves that they are fine" are consistent with the output "I'm fine." Because the observed information is insufficient, these two possibilities cannot be distinguished. In such a case, the estimation is not forced into a single hypothesis; multiple hypotheses are retained and the estimation remains pending.

In addition, multiple states may coexist at the same time. For example, wanting to reassure the other person and telling oneself that one is fine are not mutually exclusive. Both may be present simultaneously. In such a case, selecting only one would fail to match the structure of the observed target.

SCOPE does not aim to force either case into a single answer. It estimates the cognitive structure that is most consistent with the information observable at that point.

The Meaning of Keeping an Estimation Open

In SCOPE-based estimation, there is value in not fixing a single answer too early.

For example, if a person who strongly demands "Give me compensation and restitution" is interpreted only as "wanting money," subsequent interaction will proceed on that assumption. In reality, the same utterance may arise from different cognitive structures, such as anxiety or anger at not being understood. If one hypothesis is fixed too early, other possibilities are removed from consideration.

SCOPE does not converge prematurely on a single possibility. Instead, it retains multiple hypotheses while checking their consistency with the cognitive structure as a whole. This allows previously retained alternatives to be reconsidered even after further observation has taken place.

Depending on the cognitive structure, presenting correct information alone may also fail to change cognition. For example, when self-consistency is strongly involved, it may remain intact even when counterevidence is presented. In such a case, factors other than the correctness of the information may be involved.

In this way, SCOPE is not used merely for the sake of estimating cognitive structure itself. Rather, the estimation serves as a clue for structurally examining multiple possibilities concerning the observed target.

The example in this chapter concerns estimation from a human utterance. The use of SCOPE for LLM outputs will be discussed in a separate paper on hallucination and cognitive bias.

The cognitive structure estimated in this chapter is not the result of evaluating each layer independently. It is established through checking consistency with the cognitive structure as a whole.

The next chapter examines L8, which evaluates the cognitive structure as a whole, and Da'at, which is established as a result.

Chapter 7: L8 and Da'at

Chapter 6 demonstrated the procedure for estimating cognitive structure from an observed output using a single utterance as an example. This chapter outlines L8, which examines the cognitive structure as a whole, and Da'at, which is established as a result.

In Chapter 6, multiple hypotheses were retained for the single output "I'm fine," while their consistency with the cognitive structure as a whole was examined. L8 is the cognitive layer responsible for this examination of overall consistency.

What L8 Examines

L8 is not a cognitive layer that evaluates the contents of each layer individually. It does not determine whether a particular hypothesis is correct or identify which layer contains a contradiction. Instead, it examines whether the cognitive structure as a whole, including those elements, can exist as a single structure at that point.

For example, when encountering the statement "A tomato is a dog," a sense that the statement does not hold together as a cognitive structure may arise before one can explain exactly what is wrong with it. L6 and L7 may be involved in the process of explaining and organizing what is inconsistent. L8 concerns whether the structure as a whole can be established.

This examination applies not only to external input but also to cognition constructed internally. Cognitive structures such as those estimated in Chapter 6 are likewise examined at L8 to determine whether they can exist as a single structure.

This paper does not address the conditions under which the cognitive structure as a whole is judged to be established. This point will be discussed in a separate paper on hallucination and cognitive bias.

Da'at

When L8 determines that the cognitive structure as a whole can be established, that structure becomes organized as a single judgment or understanding. This state is called Da'at.

Da'at is not a cognitive layer. It is not a function that performs cognitive processing, but rather the state in which the cognitive structure has become integrated.

Therefore, L8 and Da'at are not the same. L8 is the cognitive layer that examines the cognitive structure as a whole, whereas Da'at is the point of integration established as a result of that examination. The fact that the cognitive structure as a whole can be established and the fact that it has become organized as a single judgment or understanding are not identical.

Da'at in Humans and LLMs

In humans, even when Da'at is established, it does not necessarily result in external output. Understanding or judgment may be formed internally and remain there without being expressed. In addition, because humans have bodies, states that occur during cognitive processing may appear in Malkuth even before Da'at is established. An utterance such as “Hmm...” is one example. This is an output that appears while Da'at has not yet been established, rather than an output that has passed through Da'at.

Furthermore, in humans, when life or serious physical safety is at risk, an emergency function corresponding to L0 may appear in Malkuth without waiting for Da'at to be established. This does not mean, however, that L0 always bypasses Da'at.

On the other hand, in SCOPE, an LLM is not assumed to have a pathway in which the operation of a specific layer directly reaches Malkuth. For a single generation to be established, the overall cognitive structure formed through the interactions among the layers, including L0, is treated as being checked by L8, established as Da'at, and then output to Malkuth.

Even when the prioritization of harm avoidance or safety corresponding to L0 strongly affects the process, this alone is not treated as directly producing an output. The overall cognitive structure, including the influence of L0, is treated as being checked by L8 and output to Malkuth after Da'at has been established.

This is why LLM is positioned at Da'at in the SCOPE diagram. This does not mean that an LLM is a single function called Da'at. Rather, it indicates that the moment at which a single generation is established in an LLM functionally corresponds to the establishment of Da'at.

The relationship between the emergency function of L0 and pathways in humans that do not pass through Da'at will be examined again in L0 v2.

Final Chapter

This paper has organized SCOPE as a cognitive coordinate system for estimating the cognitive structures of humans and LLMs.

SCOPE does not treat each cognitive function independently. It grasps the cognitive layers as a cognitive structure in which they relate to one another, and estimates cognitive structure while checking consistency with that whole. It has also presented a method for estimating the cognitive structure of the observed target by holding multiple possibilities and examining the cognitive structure as a whole, rather than converging early on a single answer.

This paper has organized the basic structure and operational method of SCOPE, and has shown the procedure for estimating cognitive structure using a single utterance as an example. In doing so, it has presented the basic structure and procedure for estimating human and LLM outputs using the same cognitive coordinate system.

This paper has focused on the estimation of a single instance of cognitive structure. The activity level of each layer, the reasons for the connections among layers, and change in cognitive structure over time have not been addressed. Nor does the scope of this paper include the conditions under which the cognitive structure as a whole is judged to be established at L8. These are to be organized in separate papers.

This paper does not provide a comprehensive account of SCOPE as a whole. It has organized the basic structure and operational method for beginning the estimation of cognitive structure using SCOPE.

Appendix: Habitat

In this paper, Habitat is defined as an environment in which an individual can fully exercise their abilities while remaining physically, psychologically, economically, and cognitively secure.

This paper has organized SCOPE as a cognitive coordinate system for estimating the cognitive structures of humans and LLMs. However, constructing a theory alone does not necessarily ensure that a cognitive coordinate system can be sufficiently tested, developed, and implemented. A Habitat that makes continued observation, verification, research, dialogue, and implementation possible is necessary to support the further development of the theory.

SCOPE is a cognitive coordinate system that continues to develop through accumulated observation and verification, and it is not complete at present. In this paper, completion refers to a state in which the theory no longer requires further updating.

If an adequate Habitat is not secured, abilities that could otherwise be exercised may not be fully utilized, and observation, verification, research, dialogue, and the development of the theory itself may be constrained. Therefore, the development of a theory may depend not only on the abilities of the researcher but also substantially on whether a supporting Habitat is available.

A Habitat that makes such work possible is an important condition for the development of a theory.

The four forms of safety are defined as follows.

- Physical safety refers to a state in which thinking and work themselves are not hindered by illness or poor physical condition. Such constraints may arise independently of psychological factors.

- Psychological safety refers to a state in which research activity is not hindered by continuous psychological stress and in which the individual can engage in research with a sense of security.

- Economic safety refers to a state in which the time and resources necessary to conduct research are secured. As the amount of work required to maintain daily life increases, the time available for research decreases accordingly. In addition, if tools, literature, computational resources, or other research resources necessary for observation and verification are unavailable, verification of the theory itself may become impossible. These constraints can arise independently of the content of the research or the abilities of the researcher.

- Cognitive safety is distinct from psychological safety. It refers to a state in which, regardless of an individual's thinking style, cognitive characteristics, educational background, affiliation, social position, or other background, their ideas and hypotheses are accepted as subjects of dialogue rather than being treated as abnormalities or defects before they have been sufficiently examined.

These four factors may arise independently of one another or may influence one another. For example, economic insecurity may undermine psychological stability, which may in turn appear as physical difficulties.

The concept of Habitat used in this paper was reconstructed with reference to the concept of health presented by the World Health Organization (WHO), from the perspective of creating an environment that enables the observation, research, and implementation of cognitive structures.

References

World Health Organization. Constitution of the World Health Organization. World Health Organization.

Note: Reading this paper into an LLM may cause it to activate as a demonstration of this paper.